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Digital soil mapping of key secondary soil properties using pedotransfer functions and Indian legacy soil data

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ABSTRACT

Water and nutrient holding capacities are primarily controlled by the secondary soil parameters of water retention characteristics and cation exchange capacities (CEC). Complete suite of these secondary soil parameters is rarely included in legacy soil databases generated in soil survey and mapping efforts. Therefore, pedotransfer functions (PTF) are generally used to derive these parameters. We used PTFs and laboratory-measured water contents at field capacity (θ_{FC}) and wilting points (θ_{PWP}) and CEC values to augment the existing Indian soil legacy database that contained CEC data for 1681 locations and θ_{FC} and θ_{PWP} data for about a 3rd of these locations across the country. A quantile regression forest (QRF) algorithm was used to map these three soil parameters at six GlobalSoilMap-specified soil depths. Estimated coefficient of determination (R^2) values in the 10-fold cross-validated datasets ranged from 0.60 to 0.65 for CEC while a substantially lower accuracies were found for the water holding capacities (R^2 for θ_{FC}: 0.27–0.34 and R^2 for θ_{PWP}: 0.33–0.37). Resulting soil function maps show distinct spatial patterns: high values for the water and nutrient holding parameters in soils developed from basaltic parent materials of central and part of peninsular India and low values in soils developed in the deserts of Western Plains and in soils developed from granite-gneiss parent material commonly seen in many eastern Indian region. Terrain attribute showed stronger influence on the distribution of these secondary soil parameters than the climatic parameters. Specifically, valley depth showed strongest negative correlation coefficients (r: −0.68 to −0.73) with layer-wise CEC values; January temperature showed strongest influence on water holding capacities (r: 0.08 to 0.62 with θ_{FC} and 0.14 to 0.62 for θ_{PWP}). Thus, CEC in Indian soils is primarily driven by the pedogenesis of clays in which monsoon rain favors clay translocation to deeper soil layers followed by subsequent deposition in subsurface horizons during the post-monsoon dry weathers. Similarly, strong linkages between January temperature and water holding capacities reaffirm the conclusion that temperature of driest quarters is a key driver of water relationships in soil.

1. Introduction

Soil as a multifunctional medium for plant growth is increasingly recognized as a key component of achieving sustainable development goals (Blum, 1993; European Commission, 2006; Banwart, 2011; Baveye et al., 2016; Evans et al., 2021). Soils directly contribute to 9 out of 17 sustainable development goals (Keesstra et al., 2016). Specifically, biomass production, storage and filtration of water, storage and recycling of nutrients, carbon storage and habitat for biological activity are considered major ecological functions of soils (Wiesmeier et al., 2019). Each of these functions are closely linked to the capability of a soil to sustain and increase food production – a compelling requirement in many countries including in India where farm income is mandated to double by 2022 (Chand, 2017). Improved management practices and agricultural intensification are needed to realize such goals; but, intensification has been shown to cause the deterioration in soil quality (Squire et al., 2015). Vrebos et al. (2017) observed that there is a general lack of specific strategies to address soil deteriorations in many countries. Development of a clear strategy to improve soil capability to produce more food in a sustainable manner requires that soil properties

Abbreviations: ISL, Indian soil legacy data; DSM, digital soil mapping; SOC, soil organic carbon; CEC, cation exchange capacity; SFA, soil functions assessment; PTF, pedotransfer function; NBSS&LUP, National Bureau of Soil Survey and Land Use Planning; AER, Agro-Ecological region; QRF, quantile regression forest; RF, random forest; WCLIM, World Climatic Centre; SRTM, Shuttle Radar Topography Mission.

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and functions may be quantifiable in a cost-effective and rapid manner. However, soil functions are interlinked (Blum, 1993) and several sources of uncertainty continue to plague their assessment starting from data availability to model uncertainties (Greiner et al., 2018).

Over the past few decades, several studies have been conducted to standardize soil functions assessment (SFA) methods (BBodSchG 1998; Haslmayr et al., 2016; Griener et al., 2017; Jost et al., 2021). In particular, Greiner et al. (2017) reviewed existing SFA methods and summarized key indicator variables in terms of primary and secondary soil properties. Some of the basic soil properties include soil texture, stone contents, bulk density, pH, soil organic carbon (SOC) contents, and soil hydromorphic characteristics. Secondary soil properties are generally difficult-to-measure and may be derived from basic soil properties; these properties include cation exchange capacity (CEC), volumetric water contents at field capacity (θ_{FC}) and wilting points (θ_{PWP}), hydraulic conductivity among others (Haslmayr et al., 2016; Greiner et al., 2017). Most of the basic soil properties are typically quantified as part of many national soil survey campaigns leading to the creation of rich soil legacy databases. However, conventional measurement methods for secondary soil properties are expensive and time-consuming (Romano and Palladino, 2002; Malone et al., 2020) and, therefore, most legacy soil databases often do not contain these important soil properties for all soil profiles or their individual horizons. Consequently, pedotransfer functions (PTF) are frequently used for deriving secondary soil properties from basic soil properties (Bouma, 1989; van Looy et al., 2017).

Because legacy databases provide soil data for a specific location (point data), soil mapping is done to obtain soil properties at unmeasured locations. Over the last few decades, digital soil mapping (DSM) approaches have been developed to provide soil information and inferences in a continuous manner (McBratney et al., 2003; Malone et al., 2009; Minasny et al., 2013). A GlobalSoilMap initiative has now been launched to create global maps for 12 key soil properties at six designated depths of 0–5 cm, 5–15 cm, 50–30 cm, 30–60 cm, 60–100 cm, and 100–200 cm (Arrouays et al., 2014). Concurrently, an outstanding effort is underway to compile and periodically update the World Soil Information Service (WoSIS) database (Batjes et al., 2017), which currently holds soil data for about 1.35 soil profiles per 1000 km^2 area across the world (Batjes et al., 2020). With several soil data rescue efforts underway (Arrouays et al., 2017), several regional (Dharumarajan et al., 2019) to national (Román Dobarco et al., 2019; Padarian et al., 2017; Reddy et al., 2021) and continental scale (Hengl et al., 2015) DSM studies have been reported for several basic soil properties. Similar efforts for the secondary soil properties (Table 1) are limited because of the paucity of measured data. Nevertheless, few researchers have presented DSM products for water and nutrient storage parameters at continental scale: for θ_{FC} and θ_{PWP} in Europe (Ballabio et al., 2016; Tóth et al., 2017), for CEC in Africa (Hengl et al., 2015), and for available water capacity (AWC = $\theta_{FC} - \theta_{PWP}$) and effective CEC in Australia (Grundy et al., 2015; Viscarra Rossel et al., 2015). At the national scales, digital soil maps for AWC have been developed for Scotland (Poggio et al., 2010), South Korea (Hong et al., 2013), France (Román Dobarco et al., 2019), Nigeria (Ugbaje and Reuter, 2013), Chile (Padarian et al., 2017) and Australia (Malone et al., 2020). At the regional scales and field scales, some of these secondary soil properties are reported for New

Table 1
List of some of the published digital soil mapping studies on secondary soil properties.

Sl. No	Properties	Region/Country	Area, '000 km^2	Number profiles*	Depths, cm	Prediction method	Uncertainty maps	Reference
1	θ_{FC}, θ_{PWP}, θ_s, K_s	Europe	10,180	18,000*	0–200	PTF	N/A	Tóth et al., 2017
2	SSC, coarse fragments, BD, and AWC	Europe	10,180	19857*	20	MARS	SD	Ballabio et al., 2016
3	SSC, BD, AWC, SOC, pH, effective CEC, total P and N	Australia	2,941	282,517	0–200	DTPRK	90 % CI	Viscarra-Rossel et al., 2015
4	SOC, pH, SSC, BD, CEC, TN, exchangeable acidity and bases, Al content	Africa	30,370	28,000	0–200	RF-RK	N/A	Hengl et al., 2015
5	AWC	Scotland	78	2290	N/A	GAM	90 % PI	Poggio et al., 2010
6	AWC	Nigeria	923.768	1120	0–5	Cubist	95 % PI	Ugbaje and Reuter, 2013
7	AWC	South Korea	100	380	0–100	PTF	N/A	Hong et al., 2013
8	SOC, θ_{FC}, θ_{PWP}, BD, pH, SSC	Chile	756.95	587	0–200	CART	90 % PI	Padarian et al., 2017
9	AWC	France	543.940	81,671	0–200	PTF	SD	Román Dobarco et al., 2019
10	θ_{FC}, θ_{PWP}, θ_s, AWC	Australian agricultural regions	N/A	1127	0–200	RF	half range of 95 % PI by mean	Malone et al., 2020
11	AWC	Warren and Trangie, Australia	N/A	32 and 65	0–90	MLR	N/A	Gooley et al., 2014
12	SOC, BD, SOC stock, CEC, SSC, θ_{FC}, θ_{PWP}, AWC	Parque Estadual da Mata Seca Forest, Brazil	0.102	327	0–40	OK and RK	N/A	Vasques et al., 2016
13	θ_{PWP}	Fuxin County, China	6.264	70*	0–20	BRT, SMLR	SD	Jin et al., 2018
14	SOC, pH, SSC, rock fragments EC, gypsum, $CaCO_3$, SAR, AWC, BD	Upper Colorado River Basin, USA	432	6733	0–200	QRF	95 % CI	Nauman and Duniway, 2020
15	θ_s, θ_{FC}, θ_{PWP}	Balaton catchment, Hungary	5.775	359	0–90	RF-RK	5 % and 95 % quantiles	Szabó et al., 2019
16	pH, SOC, SSC, θ_{FC}, θ_{PWP}, CEC	Northern Karnataka Plateau, India	3.655	60	0–200	QRF	95 % CI	Dharumarajan et al., 2020
17	θ_{FC}, θ_{PWP}, AWC	Narrabri, Australia	0.00023	52	0–90	SMLR	95 % CI	Zare et al., 2021
18	AWC	New South Wales	810	117*	15–30	BIM	95 % CI	Somarathna et al., 2021

MLR: multiple linear regression; SMLR: stepwise MLR; MARS: multivariate adaptive regression splines; GAM: generalized additive models; PTF: pedotransfer function; RF: random forest; OK: ordinary kriging; RK: regression kriging; CART: classification and regression tree; BRT: boosted regression tree; DTPRK: decision tree with piecewise regression and kriging; BIM: Bayesian inverse models; QRF: quantile regression forest; EC: electrical conductivity; SSC: sand, silt and clay contents; BD: bulk density; CEC: cation exchange capacity; θ_{FC}: water contents at field capacity; θ_{PWP}: water content permanent wilting point; AWC: available water capacity; θ_s: saturated water content; K_s: saturated hydraulic conductivity; CI: confidence interval; PI: prediction interval; SD: standard deviation; N/A: not available; the superscript * denotes the number of data points (number of profiles for the study was not available).

South Wales in Australia (Gooley et al., 2014; Zare et al., 2021; Somarathna et al., 2021), Upper Colorado River Basin in the USA (Nauman and Duniway, 2020), northern Karnataka Plateau in India (Dharumarajan et al., 2020) and for other countries such as in Brazil (Vasques et al., 2016), China (Jin et al., 2018), Hungary (Szabó et al., 2019). In general, estimation accuracies for water holding parameters are moderate (Nauman and Duniway, 2020) although the coefficient of determination (R^2) values as high as 0.71 to 0.77 have also been reported (Szabó et al., 2019).

<Table 1>.

A common observation in many of these studies is the sparse data availability for θ_{FC} and θ_{PWP} (Poggio et al., 2010; Hong et al., 2013; Ugbaje and Reuter, 2013; Ballabio et al., 2016; Tóth et al., 2017) leading to the use of PTFs for creating an augmented database using the existing data. Second, many of these studies provide estimates of uncertainty at various stages of mapping (Ugbaje and Reuter, 2013; Román Dobarco et al., 2019); however, uncertainties because of the use of PTFs in mapping are often not addressed. Román Dobarco et al. (2019) have shown that the prediction uncertainties in θ_{FC} and θ_{PWP} are generally lower than those obtained in the DSM validation datasets, which may be attributed to the requirement of more georeferenced data for the secondary soil properties. Thus, a major challenge in the mapping of secondary soil properties lies in the data scarcity and DSM studies in the developing countries are no exceptions.

Water and nutrient holding capacity data for Indian soils are limited with fewer datasets existing for water holding capacities than CEC. Recently, Chakraborty et al. (2020) showed that robust regional PTFs for CECs may be developed by exploiting the inherent nonlinearities and high variability in large-scale soil legacy data and the spatial structure inherent in regional datasets. Thus, opportunities exist in utilizing sparse legacy datasets for secondary soil properties. The importance of water relation studies for Indian soils is compelling. For a country with extensively irrigated agriculture (Siebert et al., 2010), distinctly depleting groundwater resources (Dangar et al., 2021), increasing frequency of extreme weather events (Singh et al., 2014) and with the predictions to have more frequent 'worst soil moisture droughts' in coming future (Aadhar and Mishra, 2021), soil moisture relationships were recognized to have greater significance in Indian soils than other parts of the world as early as 1906 (Leake, 1906). A seminal work to estimate soil hydraulic properties from soil surface area in the PTF approach dates back to 1966 (Abrol and Khosla, 1966). However, development of parametric PTFs for hydraulic properties using basic soil properties (Santra and Das 2008; Adhikary et al., 2008; Santra et al., 2018) or through reflectance spectroscopy (Santra et al., 2009) and the use of machine learning approaches for Indian soils (Jena et al., 2021) are fairly recent although most of these studies have been conducted over small geographical areas. No national-scale mapping efforts exist for understanding the nature and distribution of water holding capacities of this vast subcontinent. Similarly, no national-scale CEC maps are developed although a relatively large volume of CEC data are available in the form of published reports.

In an effort to utilize the existing soil legacy data, we recently collated an Indian soil legacy (ISL) database containing horizon-wise data on soil textural fractions, pH and SOC contents for 1707 soil profiles from published reports and publications to develop 3-D DSM products for basic soil properties at 500 m resolution for all the GlobalSoilMap-designated soil depths (Reddy et al., 2021). The objective of this study is to update the ISL database by collating CEC, θ_{FC} and θ_{PWP} data and develop national-scale digital soil maps for these three secondary soil properties at 500 m resolution for all the GlobalSoilMap-designated soil depths.

2. Materials and method

2.1. Indian soil legacy database

The ISL database (Reddy et al., 2021) contains layer-wise soil property data for 1707 georeferenced soil profiles distributed across India (latitude: 8°4′ N-37°6′ N, longitude: 68°7′ E – 97°25′ E longitudes, geographical area: 3.29 million km^2). The dataset was collated from 27 Soil Series reports published by the National Bureau of Soil Survey and Land Using Planning (NBSS&LUP), Indian Council of Agricultural Research (ICAR), New Delhi; one research bulletin from the ICAR and 11 research papers. Soil data collated from these 39 data sources (numbered 1 to 39 in Table S1) had location information in terms of specific latitude and longitude and the resulting database was referred to as the ISL database in Reddy et al. (2021). The Soil Series reports were generated as a part of the national Soil Resource Mapping (SRM) programme initiated in 1986 for developing soil maps at 1:250,000 scale for different states (Sehgal, 1990). These publications contain site characteristics, morphological features, and horizon-wise physical and chemical properties for each soil profile. Soil profiles in the ISL database are distributed over 18 of the 20 agro-ecological regions (AER) of India (Sehgal et al., 1992) although low profile density exists in several pockets of Jammu and Kashmir, Uttar Pradesh, Karnataka and some of the northeastern states of the country.

The ISL database contained horizon-wise soil data for sand contents (n = 7329), clay contents (n = 7337), soil pH (n = 7307), and SOC contents (n = 7280). For the current study, we expanded this dataset by collating horizon-wise data for CEC (n = 7198), θ_{FC} (n = 2898), and θ_{PWP} (n = 2901) data. While the CEC data are available for 1681 soil profiles, the $_{FC}$ and θ_{PWP} data are available only for 774 and 775 profiles, respectively (shown by the circles with inscribed '+' symbols in Fig. 1; because 1,681 profiles had CEC data, we have not indicated those 26 locations for which CEC data was not available in Fig. 1). Specifically, the eastern parts of Indo-Gangetic Plains (AER 9 and 14) and the northeastern hilly regions (AER 13 to 17) did not have any θ_{FC} and θ_{PWP} data; these AERs occupy a total area of 0.767 million km^2 adjoining the Himalayan mountain ranges. Fig. 1 also shows that most of the soil profiles containing θ_{FC} and θ_{PWP} data are distributed over the western and central parts of India covering about 2.317 million km^2 (~77 % of total geographical area) in 12 different AERs. Such a distribution yields about 0.33 profiles per 1000 km^2, which is comparable to some of the national-scale DSM studies (e.g., China: Song et al., 2020; Australia: Malone et al., 2020) but are much lower density than the others (e.g., France; Mulder et al., 2016).

Only fewer soil profiles from the ISL database (Reddy et al., 2021) contained θ_{FC} and θ_{PWP} data. Hence, we augmented this ISL database with soil data from other sources that contained CEC, θ_{FC} and θ_{PWP} data but did not have locational information (serial number 40 to 77 in Table S1. With these data sources, the augmented ISL database contained CEC data for 10,087 horizons, θ_{FC} data for 3407 horizons and θ_{PWP} data for 3132 horizons along with 5 basic soil properties (i.e., sand, silt, clay, and SOC contents and soil pH). Although the addition of these data sources slightly increased the number of θ_{FC} and θ_{PWP} records, many of these new data did not have corresponding location (latitude and longitude) information except for the name of the sampling sites. Thus, the total number of georeferenced soil profile locations remained unaltered at 1707 for DSM modelling. However, the PTFs developed with the combined soil horizon data (i.e., soil horizon data both with and without location information) performed better than those developed with known latitude and longitude information. Detailed sources of soil data and area-wise distribution of AERs may be seen in the Supplementary Materials (Table S1 and Table S2).

2.2. Dependency analysis and PTF development

Before developing PTFs, both Pearson and distance correlation

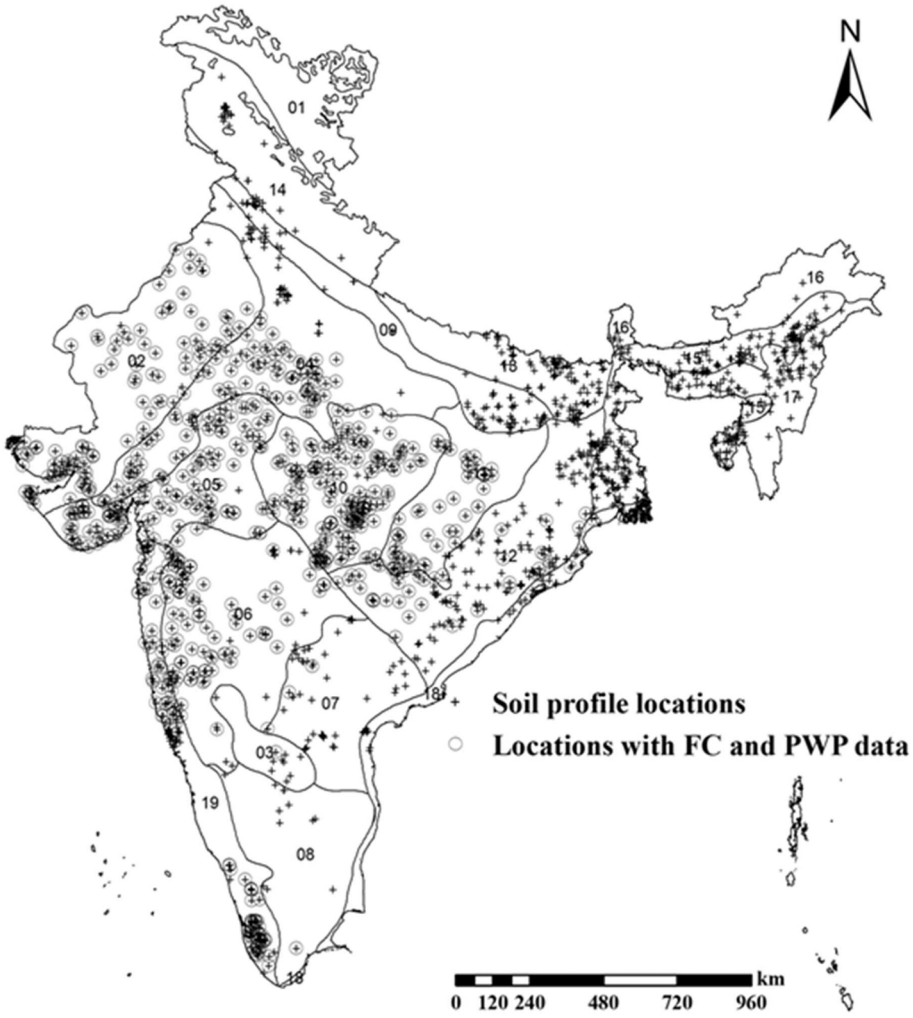


Fig. 1. Map showing locations for soil profiles and boundaries of the agroecological regions (AER) in India. Numbers indicate the specific AERs. The symbol '+' indicates the soil profile locations (Reddy et al., 2021). Circles around '+' symbols indicate the locations for which water contents at field capacity (θ_{FC}) and wilting point (θ_{PWP}) data were available. The map and AER boundaries were obtained by digitizing the AER map of Sehgal et al. (1992).

(dCor) analyses (Székely et al., 2007) were undertaken to examine the relationships between each of CEC, θ_{FC} and θ_{PWP} and their predictor variables such as basic soil properties and environmental covariates. The dCor is a nonlinear dependency measure based on distribution or density functions and is given as:

$$\mathrm{dCor} = \frac{dcov(x,y)}{\sqrt{dcov(x,x)dcov(y,y)}} \tag{1}$$

where x and y are two random vectors and *d*cov is the distance covariance. The *d*cov is given as:

$$dcov^2(\mathrm{x},\mathrm{y}) = \int_{R^{dx+dy}} \left\|\phi_{\mathrm{x,y}}(t,s) - \phi_{\mathrm{x}}(t)\phi_{\mathrm{y}}(s)\right\|^2 w(t,s)dtds \tag{2}$$

where $\phi_x(t)$ and $\phi_y(s)$ are the respective characteristic functions of the random vectors × and y, $\phi_{x,y}$(t, *s*) is the joint characteristic function of × and y and d_x and d_y are the dimensions of × and y, *w*(t, *s*) is the weight function and is given by

$$w(t,s) = \left\{c_{d_x}c_{d_y}\|t\|_{d_x}^{1+d_x}\|s\|_{d_y}^{1+d_y}\right\}^{-1} \tag{3}$$

where$c_d = \pi^{(1+d)/2}/\Gamma\{(1+d)/2\}$. The dCor value between two variables is zero if and only if the variables are independent (Székely et al., 2007).

Preliminary modelling studies using multiple linear regression (MLR) and several machine learning approaches showed that secondary soil properties may be readily estimated using the random forest (RF; Breiman, 2001) and extreme gradient boosting (XGB) algorithms (Chen and Guestrin, 2016). The RF model follows a bagging algorithm in which each node is split using the best among a subset of predictors randomly chosen at that node following a bootstrap approach (Breiman, 2001). The XGB algorithm involves an ensembling technique where we build many independent models and combine them using model averaging technique (Chen and Guestrin, 2016). We used the *randomforest* and *xgboost* to implement the above modelling approaches in the R software (R Core Team, 2019) and RStudio 1.0.153 (RStudio Team, 2015). For the predictor variables, we tested basic soil properties and different environmental covariates (Table S3) in multiple combinations. Results showed that the addition of environmental covariates to the basic soil properties did not improve the performance of the developed PTFs. Chakraborty et al. (2020) also obtained similar result for CEC. Moreover, the MLR approach did not perform well in comparison with machine learning approaches. Therefore, we considered only the basic soil properties (soil textural fractions, SOC contents and soil pH) as predictor variables in the RF and XGB algorithms for estimating CEC, θ_{FC} and θ_{PWP} data for the soil horizons for which these three properties were missing. Table 2 shows the 10-fold cross validation statistics for these secondary soil properties using RF and XGB algorithms, which shows that the RF model performs similarly to a greedy algorithm such as XGB. For the

Table 2
Coefficient of determination (R^2) and root-mean-squared error (RMSE) values for cation exchange capacity (CEC), field capacity (θ_{FC}) and wilting point (θ_{PWP}) estimated using the random forest (RF) and extreme-gradient boosting (XGB) algorithms.

Property	N	Random forest		Extreme-gradient boosting	
		R^2	RMSE	R^2	RMSE
CEC ($cmol_c\ kg^{-1}$)	10,087	0.76	7.20	0.75	7.27
θ_{FC} (%)	3407	0.67	6.2	0.65	6.36
θ_{PWP} (%)	3132	0.61	4.0	0.60	4.15

N: total number of samples used in RF and XGB modelling.

saturated hydraulic conductivity, Jena et al. (2021) recently showed that RF approach performed better than other machine learning approaches for developing PTFs. Several DSM studies also have shown RF may be an effective machine learning algorithm (Taghizadeh-Mehrjardi et al., 2021). Hence, for the remainder of our study, we chose to use only RF algorithm to estimate CEC, θ_{FC}, and θ_{PWP} data for all the 1707 locations. After the application of PTFs, a common dataset containing 7331 soil layers for CEC, 7273 soil layers for θ_{FC}, and 7736 soil layers for θ_{PWP} values were generated for DSM modelling.

We also conducted a scenario analysis to examine the extent of error that propagates to the final DSM outputs through developed PTFs. Following a similar approach to Ramirez-Lopez et al. (2019), we created three scenarios using the CEC dataset for which measured legacy data were available for 1681 out of total 1707 soil profile locations. For the scenario 1, we used these legacy CEC data for all the 1681 profile locations and developed DSM products for each of the six soil depths. The resulting DSM maps from this scenario contains no error component because of PTFs because no PTF is used in this case. For scenario 2, we extracted a subset of CEC data from scenario for which θ_{FC} and θ_{PWP} data were available in the legacy dataset. We had legacy soil data for θ_{FC} and θ_{PWP} data for only about 773 profile locations. Many of these profiles did not have θ_{FC} and θ_{PWP} data for all the six soil horizons. To build robust PTFs, we expanded the original legacy database of Reddy et al. (2021) by adding soil data generated in our laboratory and those available from the published literature. Thus, we added non-legacy CEC data (n = 4809) to the available legacy CEC data in respective soil layers for developing PTFs for CEC similar to θ_{FC} and $\boldsymbol{\theta_{PWP}}$. Resulting PTFs were then used to estimate CEC for all the 1681 profile locations. Estimated CEC values were used in the QRF approach for developing DSM for CEC. Thus, the difference between the DSM maps from scenario 2 and scenario 1 should be the extent of error that resulted because we used the PTF to estimate CEC for all the locations in our study. For scenario 3, we developed the PTF for CEC as in the case of scenario 2. However, instead of using the PTF for estimating CEC for all the 1681 locations, we retained the measured legacy data for the 773 locations; but we estimated the CEC values for the remaining 908 locations (=1681 – 773) using the developed PTF for CEC. The combined data were then used for DSM mapping using the QRF approach. Thus, our scenario 3 is what exactly was done for missing θ_{FC} and θ_{PWP} data in the 1707 profile dataset. Resulting DSM products for these three scenarios were compared to evaluate the error propagation because of PTF models to the final DSM outputs.

2.3. Selection of environmental covariates and digital soil mapping

We used soil attributes (*s*), climate (*c*), relief (*r*) and spatial position (*n*) of the *scorpan* model (McBratney et al., 2003). The average monthly temperature and precipitation values were collected from the WorldClim – Global Climate Data (Fick and Hijmans, 2017) as climatic covariates (data available from 1970 to 2000 at 1 km^2 spatial resolution). Digital elevation model (DEM) data from the 90 m shuttle radar topographic mission (SRTM) were used to derive 15 different covariates using the System for Automated Geoscientific Analyses (SAGA v.6.3.0) software. We also used the MODIS satellite data taken over 2000–2020 to estimate the mean monthly normalized difference vegetation index (NDVI) and the enhanced vegetation index (EVI) values using the Google Earth Engine. Thus, we derived 63 different covariates:15 terrain variables from the SRTM data, 24 climatic variables from the WorldClim data; 24 vegetation indices (12 NDVI and 12 EVI values) from the MODIS data (Table S3). All these covariates were resampled to 500 m resolution for the whole country using the bilinear interpolation method built in the *raster* package in R software.

Because of large number, we used a hybrid feature selection algorithm consisting of filtering and wrapping steps to select an optimal set of covariates for DSM modeling. For the filtering step, the Kendall's rank correlation coefficient (τ) was used to select significant covariates (predictor variables) based on the *p*-value < 0.05. The wrapper (also called as Boruta) algorithm is built around the RF classification algorithm (Kursa and Rudnicki, 2010), which further reduces the number of significant covariates based on the maximum *Z*-score of shadow attributes (MZSA). The hybrid feature selection algorithm was run in the R program using the *Boruta* package. Although we used 1000 iterations in Boruta and the *p*-value was reduced to 0.001 from 0.05, as many as 53, 39, and 48 covariates were retained for CEC, θ_{FC}, and θ_{PWP}, respectively, in the feature selection process with the June rainfall being the only common covariate among the top 5 significant covariates for all the three properties. To reduce overfitting, we manually examined the performance of DSM by successively adding 5 Boruta-selected covariates to an initial set of top 5 significant covariates. These preliminary modeling results showed that 15–20 top covariates from Boruta are sufficient to develop DSM products (data not shown). Therefore, we used the top 20 significant covariates (Table S4) for the respective soil properties from the hybrid feature selection algorithm in the final DSM modeling.

We used quantile regression forest (QRF; Meinshausen, 2006) approach (Vaysse and Lagacherie, 2017; Dharumarajan et al., 2021; Hounkpatin et al., 2022; Žížala et al., 2022) to derive DSM products for CEC, θ_{FC}, and θ_{PWP}. Each of these secondary soil properties were mapped at six standard depths (0–5, 5–15, 15–30, 30–60, 60–100 and 100–200 cm) using the harmonized soil properties and Boruta-selected covariates as inputs in the QRF model. The QRF algorithm is an extension of the RF, except that in case of RF, only the conditional mean of the modelled soil property for each node in each tree is preserved, whereas the QRF approach preserves both the mean and the full conditional distribution for the modeled soil property (Meinshausen, 2006; Vaysse and Lagacherie, 2017). The number of trees was set to 500 and the number of variables used to grow each tree (*mtry*) was obtained using the random search approach. A 10-fold cross-validation with five repetitions was performed using the R *caret* package. Performance of DSM models was compared using the root-mean-squared error (RMSE) and R^2 and mean error (ME) statistics:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(Y_i - \widehat{Y}_i)^2} \quad (4)$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(Y_i - \widehat{Y}_i)^2}{\sum_{i=1}^{N}(Y_i - \overline{Y}_i)^2} \quad (5)$$

$$Mean\ Error = \frac{\sum_{i=1}^{N}(\widehat{Y}_i - Y_i)}{N} \quad (6)$$

where N is the number of observations, Y_i and $\hat{Y}_i$ are the observed and predicted secondary soil properties for the ith location, respectively, and $\overline{Y}$ is the mean of the observed soil property. The prediction uncertainty of the maps was estimated by mapping the lower (5 %) and upper (95 %) limits of the 90 % prediction interval using the QRF approach. The final product consisted of 18 mean maps for three secondary soil properties (each for six different depths) and their corresponding uncertainty maps.

2.4. Area under cultivation and irrigation, fertilizer use, and grain yield

In addition to ISL data, we collected state-wise average data for selected agronomic attributes (GSA: gross sown area, NSA: net sown area, GIA: gross irrigated area, NIA: net irrigated area, food grain yield, NPK fertilizer consumption, and CI: cropping intensity) available for 2004 to 2017 in The Reserve Bank of India website (RBI, 2020). Both gross and net irrigation intensities (GII and NII) were calculated by dividing the respective irrigated areas with the cropped areas and all the agronomic attributes were further averaged over 14-year time period. Because AER-wise agronomic data was not available, we extracted soil properties from the DSM products for each state to obtain relationships between soil properties and agronomic attributes.

3. Results and discussions

3.1. Measured and derived water and nutrient holding capacities of Indian soils

Table S5 shows the descriptive statistics for measured CEC, θ_{FC} and θ_{PWP} values collated from three different sources. The NBSS&LUP data are obtained from soil survey across India while two other sources cover relatively small geographical areas in respective study sites. This is reflected in the narrow range in all the three soil properties from the IITKgp database. However, the mean and SD values for all the properties are comparable across three data sources except for slightly lower SD values for water holding capacities in the IITKgp database. Thus, we used these combined datasets to develop PTFs with reasonably high validation R^2 values (Table 2) such that an augmented ISL database containing all the 1707 soil profile data could be compiled with the addition of PTF-derived CEC, θ_{FC} and θ_{PWP} values in addition to already existing basic soil properties (sand and clay contents, SOC contents, and soil pH) used in DSM modelling earlier (Reddy et al., 2021). Although we used an augmented dataset for PTF development with efficient machine learning approaches to fill missing data in the ISL database, measured soil hydraulic properties for all the profiles are needed to avoid any error propagation because of the PTF estimati$_\theta$n into the estimation of soil properties in the DSM approach.

Descriptive statistics for both the measured and PTF-augmented CEC, θ_{FC} and θ_{PWP} values are listed in Table 3, which shows that the CEC values in Indian soils range from 0.5 to 96.0 $cmol_c\ kg^{-1}$ with a median value of 14.3 $cmol_c\ kg^{-1}$. Similarly, the θ_{FC} and θ_{PWP} values range from 2.5 to 65.5% and 1.2 to 48.5% with the mean values of 27.1 and 14.1%, respectively. Resulting coefficient of variation (CV) for CEC was large (CV = 77%) whereas θ_{FC} and θ_{PWP} values moderately varied with CV values of 35% and 40%, respectively. Reddy et al. (2021) estimated 25th and 75th percentiles for the basic soil properties (sand content: 18–54.1%, clay contents: 21.9–47.7%, SOC contents: 0.20–0.66%, soil pH: 5.8–7.9) showing a wide range in soil properties for the Indian soils.

Table 3
Descriptive statistics for cation exchange capacity (CEC), field capacity (θ_{FC}) and wilting point (θ_{PWP}) in the measured datasets and after missing data were replaced using pedotransfer functions (PTFs).

Attributes	CEC ($cmol_c\ kg^{-1}$)	θ_{FC} (%)	θ_{PWP} (%)
A. Measured secondary soil function parameters			
Sites	1681	774	775
N	7198	2898	2901
Mean	19.6	26.8	14.0
Median	14.1	26.5	13.5
SD	15.13	11.01	6.90
Min	0.5	2.5	1.2
Max	96	65.5	48.5
B. Secondary soil function parameters after missing data are replaced using PTFs			
Sites	1706	1701	1701
N	7331	7273	7736
Mean	19.8	27.1	14.1
Median	14.3	27.5	13.6
SD	15.2	9.4	5.6
Min	0.5	2.5	1.2
Max	96.0	65.5	48.5

Although DSM products provide soil properties at desired soil depths for each pixel, variations of harmonized CEC, θ_{FC} and θ_{PWP} values for the soil profile locations with soil depths are shown in Fig. 2 with their descriptive statistics listed in Table S6. The mean values of all the three properties increase with soil depth. Reddy et al. (2021) reported that the clay contents and pH values in Indian soils increase with soil depth. The CEC values followed the similar trend as clay contents (Fig. 2), which is expected because CEC is strongly influenced by the charge characteristics of clay contents. Significant positive correlation ($r = 0.73$, dCor = 0.71) between CEC and clay contents (Table 4) further supports this observation. Similar to CEC, the θ_{FC} and θ_{PWP} values also showed positive correlation with clay contents. Sand contents showed the maximum negative correlation coefficients with both the water holding capacities suggesting that sand contents primarily control water relations while clay contents control CEC variations in Indian soils.

The median CEC contents of about 14.3 $cmol_c\ kg^{-1}$ (Table 3) suggest that bulk of Indian soils may possibly be at an advanced stage of weathering with the dominance of kaolinitic clays (CEC: 5–20 $cmol_c\ kg^{-1}$) barring black cotton soils (Vertisols) of the central Indian and Peninsular region. Recently, Chowdhury et al. (2021) suggested that the Singhbhum Craton in the eastern Indian region may be the oldest landmass on earth. High rainfall and temperatures in such old rock systems may be favoring high weathering intensities to have 'mature soils' with low CEC and low water holding capacities (Table 3) for much of the eastern Indian soils.

3.2. Digital soil mapping of CEC, θ_{FC} and θ_{PWP}

Table 5 lists the performance statistics for the cross validated RF model for CEC, θ_{FC}, and θ_{PWP} at six different depths. The CEC showed better predictions across six depths among the three soil properties with the R^2 and RMSE values ranging from 0.60 to 0.65 and 8.96–9.59, respectively. Similar results were reported for CEC by other researchers (e.g., Mulder et al., 2016: R^2 of 0.11 to 0.24; Chagas et al., 2018: R^2 = 0.47; Dharumarajan et al., 2020: R^2 of 0.31 to 0.56; Angelini et al., 2017: R^2 of 0.21 to 0.53). The R^2 values range between 0.27 and 0.34 and 0.33–0.37 for θ_{FC} and θ_{PWP}, respectively, whereas the RMSE values range between 7.45 and 8.24 for θ_{FC} and 4.34–4.57 for θ_{PWP}. Similar results were reported by other researchers. For example, Padarian et al. (2017) reported R^2 values ranging from 0.17 to 0.33 and 0.17–0.30 for θ_{FC} and θ_{PWP}, respectively. Román Dobarco et al. (2019) reported R^2 values of 0.21 and 0.29 for $_{FC}$ and θ_{PWP}, respectively. However, Dharumarajan et al. (2020) reported higher R^2 values ranging from 0.30 to 0.49 for θ_{FC} and θ_{PWP} at six depths for the Northern Karnataka Plateau region of the country. All the predictions of the three soil properties were unbiased with mean error values close to zero (Table 5).

The DSM products for the mean CEC, θ_{FC} and θ_{PWP} estimates and their lower (5 %) and upper (95 %) limits of 90 % prediction interval at 0–5, 5–15 and 15–30 cm soil depths are shown in Fig. 3, Fig. 4 and Fig. 5, respectively. The maps for the remaining three soil depths (30–60, 60–100 and 100–200 cm) are shown in the Supplementary Materials (Fig. S1 to Fig. S3). The summary statistics for the predicted maps are listed in Table S7 (Supplementary Materials) which shows that the mean values of CEC, θ_{FC} and θ_{PWP} increase with soil depth. Most of the higher values of CEC fall in the central and western part of the country (specifically, parts of Gujarat), which are Vertisols dominated by montmorillonitic clays (Virmani et al., 1982). Reddy et al. (2021) reported higher vθlues of clay contents, pH, and SOC contents which influence CEC values of soils in this region. Also, the region generally tends to have higher water holding capacity values (Virmani et al., 1982). The Western Ghats and eastern parts of the country are mostly acidic in nature

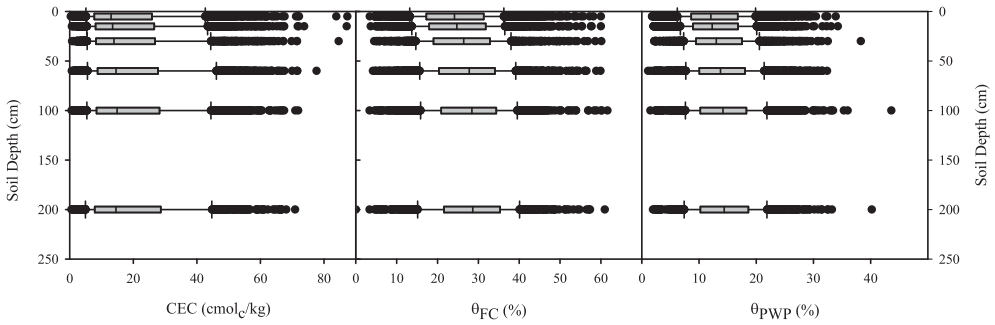


Fig. 2. Harmonized cation exchange capacities (CEC) and volumetric water contents at field capacity (θ_{FC}) and wilting point (θ_{PWP}) at 0–5 cm, 5–15 cm, 15–30 cm, 30–60 cm, 60–100 cm, 100–200 cm soil depth.

Table 4
Pearson correlation coefficients (unitalicized data) and distance correlation coefficients (italicized data) between primary and secondary soil parameters for Indian soils.

AER #	Sand %	Clay %	pH	SOC %	CEC cmol$_c$ kg^{-1}	θ_{FC} %	θ_{PWP} %
Sand, %	1.00	*0.70*	*0.15*	*0.15*	*0.52*	***0.85***	***0.67***
Clay, %	−0.74	1.00	*0.15*	*0.19*	***0.71***	*0.67*	*0.62*
pH	−0.07	0.06	1.00	*0.36*	*0.44*	*0.13*	*0.14*
SOC, %	−0.10	0.11	−0.34	1.00	*0.15*	*0.10*	*0.18*
CEC, cmol$_c$ kg^{-1}	−0.53	**0.73**	0.42	−0.01	1.00	*0.50*	*0.39*
θ_{FC}, %	**−0.85**	0.72	0.09	0.04	0.55	1.00	*0.71*
θ_{PWP}, %	**−0.65**	0.62	−0.11	0.15	0.42	0.67	1.00

SOC: Soil organic carbon content; CEC: cation exchange capacity; θ_{FC}: volumetric water content at field capacity; θ_{PWP}: volumetric water content at permanent wilting point.

Table 5
Coefficient of determination (R^2), root-mean-squared error (RMSE) and mean error (ME) values for cation exchange capacity, volumetric water content at field capacity (θ_{FC}), and volumetric water content at permanent wilting points (θ_{PWP}) estimated using random forest modelling approach.

Soil properties	Soil depth, cm	R^2	RMSE	ME
CEC (cmol$_c$ kg^{-1})	0–5	0.60	9.47	0.059
	5–15	0.62	9.19	0.071
	15–30	0.65	8.96	0.055
	30–60	0.65	9.09	0.032
	60–100	0.65	8.98	0.046
	100–200	0.61	9.59	0.102
θ_{FC} (%)	0–5	0.33	7.65	0.057
	5–15	0.34	7.45	0.073
	15–30	0.33	7.49	0.046
	30–60	0.31	7.64	0.061
	60–100	0.29	7.86	0.051
	100–200	0.27	8.24	0.004
θ_{PWP} (%)	0–5	0.33	4.49	0.053
	5–15	0.34	4.36	0.047
	15–30	0.35	4.37	0.049
	30–60	0.37	4.34	0.052
	60–100	0.35	4.53	0.037
	100–200	0.37	4.57	0.018

and receive high annual rainfall. The presence of the illites, smectite and kaolinite type of secondary silicates (Mohanty et al., 2016) because of the intense weathering and leaching results in lower CEC values (Bhattacharyya et al., 2010; Jena et al., 2016) in these regions (Fig. 3). Higher sand content in the western parts of the country may have contributed to lower θ_C and θ_{PWP} values in western Indian desert ecosystems (Fig. 4 and Fig. 5). The subsurface soils exhibit higher θ_{FC} and θ_{PWP} following the clay distribution (Srinivasarao et al., 2009). High-water contents at PWP may be attributed to the presence of higher smectitic clay minerals in vertisols (Diwakar and Singh, 1994). Overall, the amount and nature of soil textural fractions in the regions cause variations in water retention properties (Diwakar and Singh, 1994; Walia et al.,1999; Srivastava et al., 1998; Santra et al., 2018).

Results of the 90 % prediction intervals (Figs. S4-S6) show that estimated intervals are generally wide for all the three soil properties. Specifically, narrow prediction ranges may be seen in the regions where the sampling density is high and wide ranges may be seen along the southern part of the country where the sampling density is low. The uncertainty may be minimized by increasing the sampling density across the country. We also conducted a simple scenario analysis to evaluate the extent of error caused by missing θ_{FC} and $\boldsymbol{\theta_{PWP}}$ data in our 1707 legacy database (Fig. 6). With the available legacy CEC data for 1681 profile locations, scenario 1 provided DSM estimates of CEC at each profile locations; thus, the RMSE values (4.42 to 4.75 cmol$_c$ kg^{-1}) estimated using observed and DSM-estimated CEC values for these 1681 locations represent the extent of error because of DSM model itself. Estimated RMSE values for scenario 2 (4.23–4.52 cmol$_c$ kg^{-1}) suggest that the combination of PTF and DSM reduced the total error. The RMSE values of 4.48–4.66 cmol$_c$ kg^{-1} from the scenario 3 (the approach followed for estimating missing θ_{FC} and θ_{PWP} values in the legacy dataset in our study) lie in between those for scenario 2 and scenario 1. These results suggest that the use of PTF may actually reduce the total error during DSM modelling. Nevertheless, we emphasize that the creation of legacy data with high soil profile density is a prime requirement for decreasing the 90 % prediction interval in for estimating soil properties using the DSM approach.

3.3. Secondary soil properties and their relationship with basic soil properties

Depth-weighted average basic soil property data for different AERs were presented earlier (Reddy et al., 2021). For ready reference, here we list the average values of both basic soil properties and secondary soil properties for six GlobalSoilMap-specified soil depths for each AERs (Table S8). Important trends and associations among these soil properties may be seen from this Table. As was noted in Reddy et al. (2021), soils in the Deccan Plateau (AER6) have the highest clay contents and those of Western Ghats & Coastal Plains (AER 19) have the highest SOC contents among 20 different AERs. For soil pH, soils in the North Eastern Hills of Purvachal (AER 17) were most acidic while those of the Western Plains (AER 2) showed most alkaline pH values. As is expected, the AER-averaged sand and clay contents are inversely related to each other with almost all the AERs showing an increasing trend in the clay contents to a depth of 60 cm to 100 cm from soil surface. Such an increase is because of the generally high weathering intensities caused by high temperature

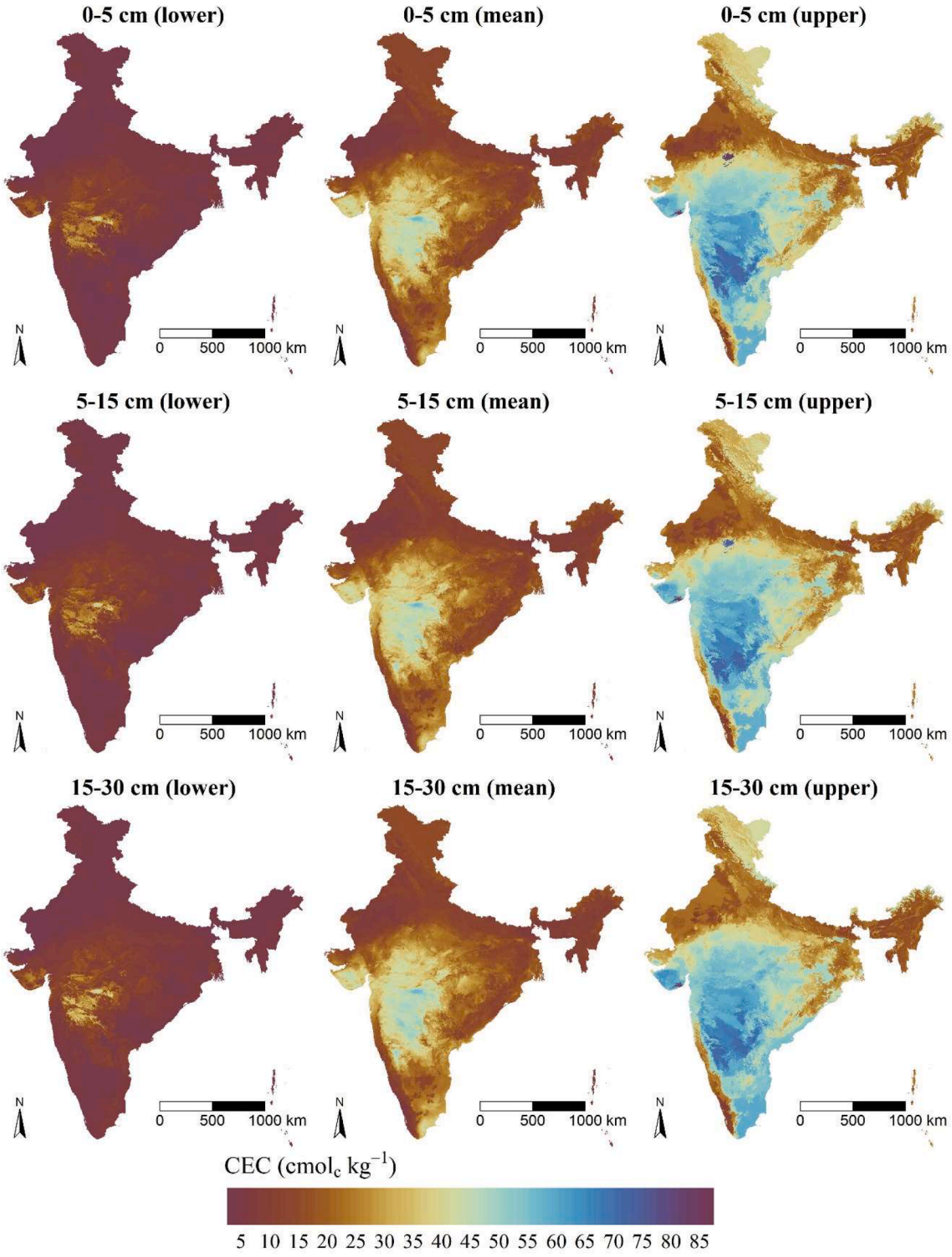


Fig. 3. Digital maps for cation exchange capacities (cmol$_c$ kg^{-1}) at 0–5, 5–15 and 15–30 cm soil depths and their lower (5 %) and upper (95 %) quantiles calculated using QRF approach.

and high rainfall observed in different AERs and consequent clay translocation to deeper soil layers. In contrast, SOC contents within a profile generally decreased exponentially with soil depth and such soil characteristics are seen for all the AERs suggesting that SOC translocation to lower soil layers is less conspicuous than soil clays for the Indian soils. Lower SOC contents in deeper soils may also be a result of overall low SOC contents of Indian soils. Reddy et al. (2021) reported a strong positive association between the whole profile clay contents and the whole profile SOC contents. Similar trends also may be seen in depth-wise *r* values between soil clays and SOC contents.

Depth-averaged CEC values for the Indian AERs ranged from 9.9 cmol$_c$ kg^{-1} for the North Eastern Hills of Purvachal (AER 17) to 41.6 cmol$_c$ kg^{-1} for the Deccan Plateau (AER 6). Presence of kaolinitic clays (Ghosh et al., 1974) in granite-gniess formations of eastern Indian regions accounts for the low CEC values in those soils while the smectitic clays (Bhattacharyya et al., 2013) in the basaltic parent material of

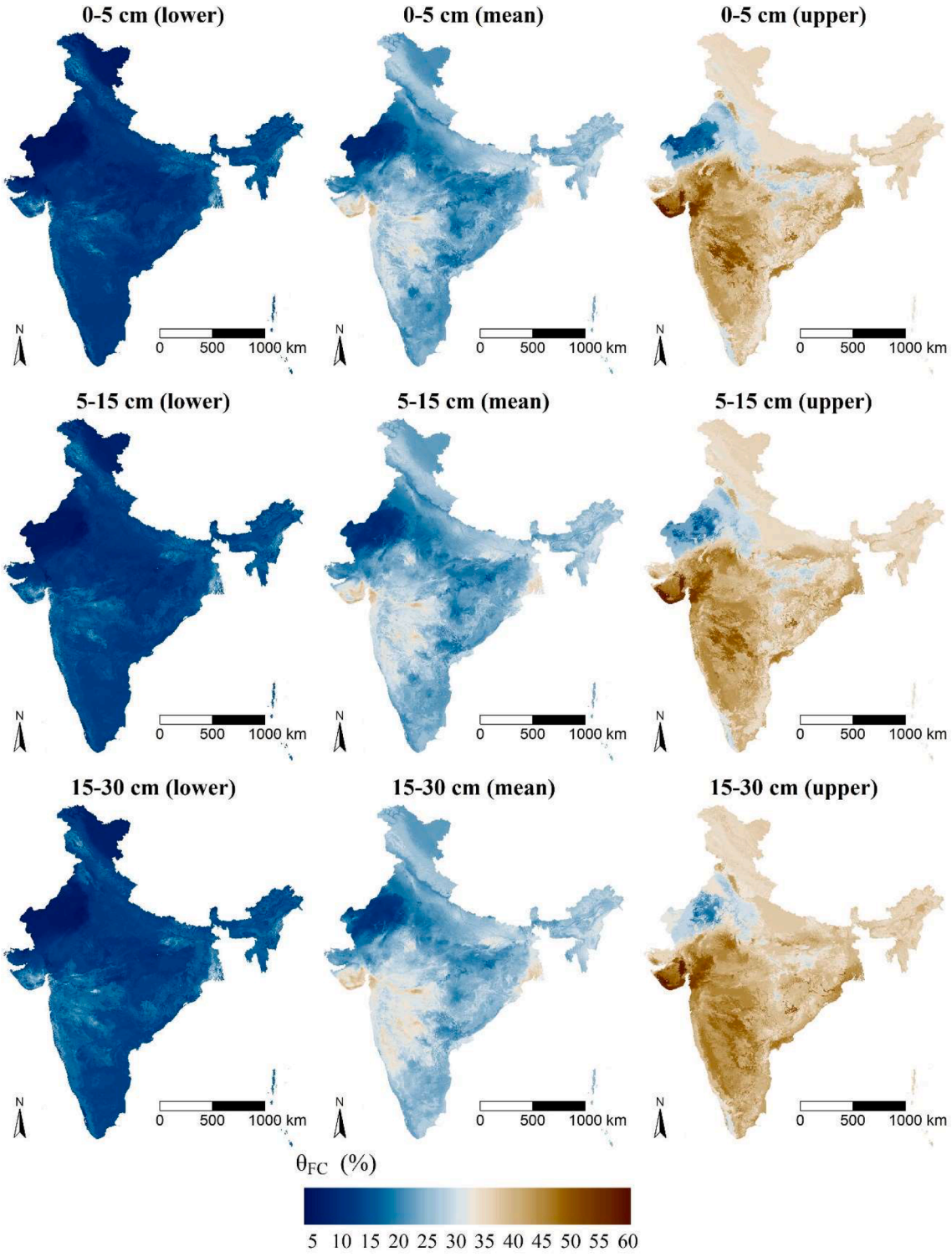


Fig. 4. Digital maps for field capacity (%) at 0–5, 5–15 and 15–30 cm soil depths and their lower (5 %) and upper (95 %) quantiles calculated using QRF approach.

Dharwar and Aravalli–Bundelkhand cratons (De et al., 2016) account for the high CECs in the central Indian soils. The increasing trend of CEC with soil depth primarily mimics the increasing clay contents for the majority of Indian soils as is evident from high *r* values (ranging from 0.76 to 0.87) between AER-averaged CEC and clay contents. Corresponding *r* values ranged from −0.12 to −0.59 for sand, and 0.42 to 0.55 for CEC with soil pH. Interestingly, CEC values showed negative correlation (*r* ranged from −0.02 to −0.30) with the SOC values for the top 30 cm soil depths (0–5 cm, 5–15 cm, and 15–30 cm) and positive correlation (*r* values of 0.05 to 0.26) with those of deeper soil layers. For all the AERs, the SOC contents exponentially decrease with soil depth; the sharp decline in the SOC contents in the top 50 cm soil layer may be responsible for the negative correlation between the clay-controlled increasing CEC values with soil depth for the top 30 cm soil layers. By spatially sub-sampling CEC data and through both linear and nonlinear correlation analysis, Chakraborty et al. (2020) observed that CEC values to have the strongest non-linear correlation with clay content; the strength of correlation decreases in the order of clay contents > pH >

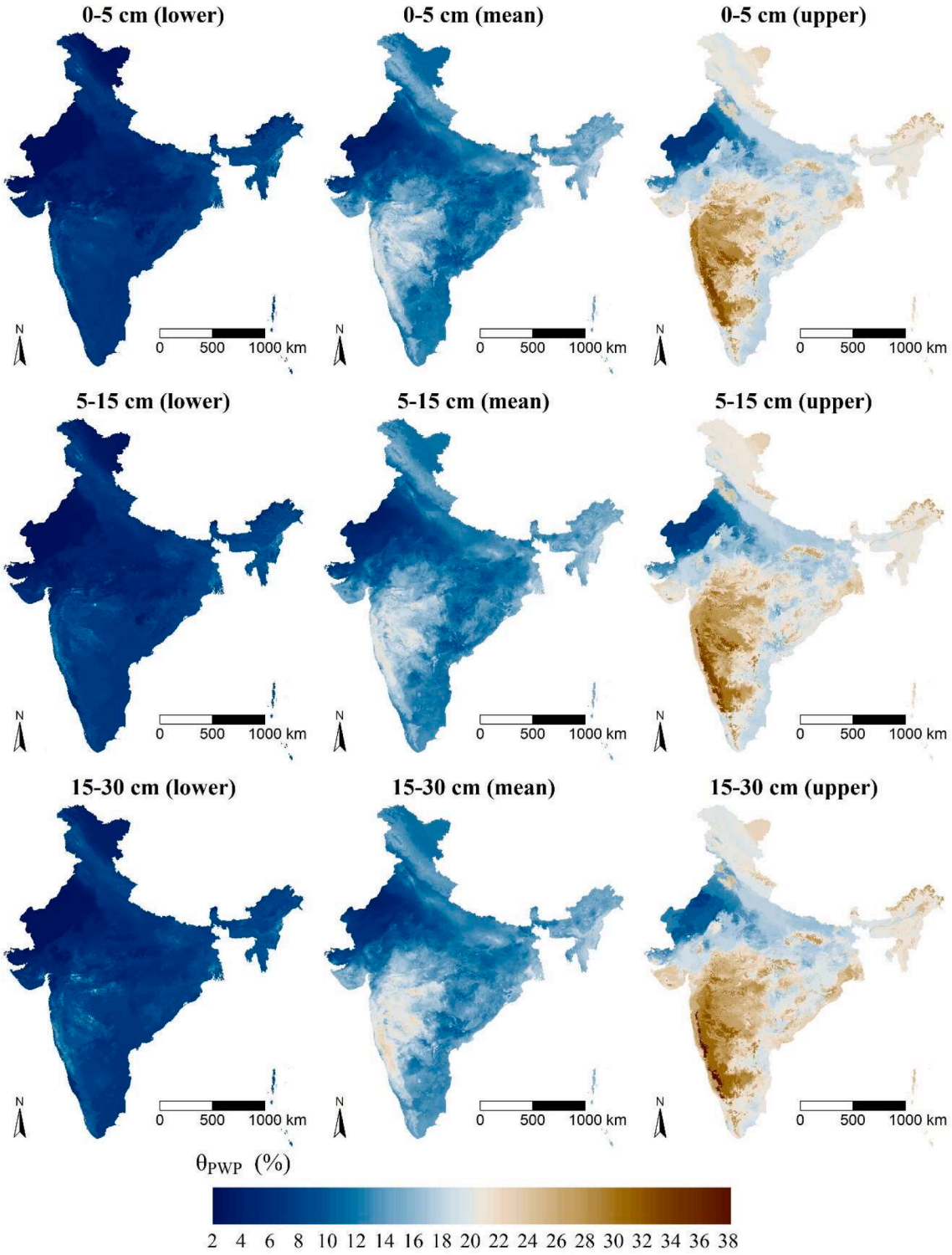


Fig. 5. Digital maps for permanent wilting point (%) at 0–5, 5–15 and 15–30 cm soil depths and their lower (5 %) and upper (95 %) quantiles calculated using QRF approach.

SOC contents. A quick reckoning of CEC data in Table S2 would reveal that soils in the majority of Peninsular and Central India covering the Deccan Plateau (AER 6, 3, and 7), Central Highlands (AER 5, 10), and Eastern Plateau (AER 11) have high CEC values (>20 $cmol_c\ kg^{-1}$) and the remaining 14 AERs (~67 % of total geographical area) have CEC values <20 $cmol_c\ kg^{-1}$. Inspection of raw data for different soil horizons show that 4632 soil horizons out of the total 7139 soil horizons (i.e., ~65 % of raw data) had CEC values <20 $cmol_c\ kg^{-1}$. Thus, the nutrient holding capacity of the bulk of Indian soils should be considered low, which should be a compelling reason for initiating specific steps for increasing the SOC contents of these soils.

The DSM-estimated average θ_{FC} values ranged from 14.6 % for the 0–5 cm soil layers of the Western Plains (AER 2) to 35.8 % for the 100–200 cm soil layers of the Deccan Plateau (AER 6). Similarly, the surface soil layer in the AER 2 showed the minimum PWP value of 6.1 % and the 30–60 cm layer in AER 6 showed the highest PWP value of 20.2

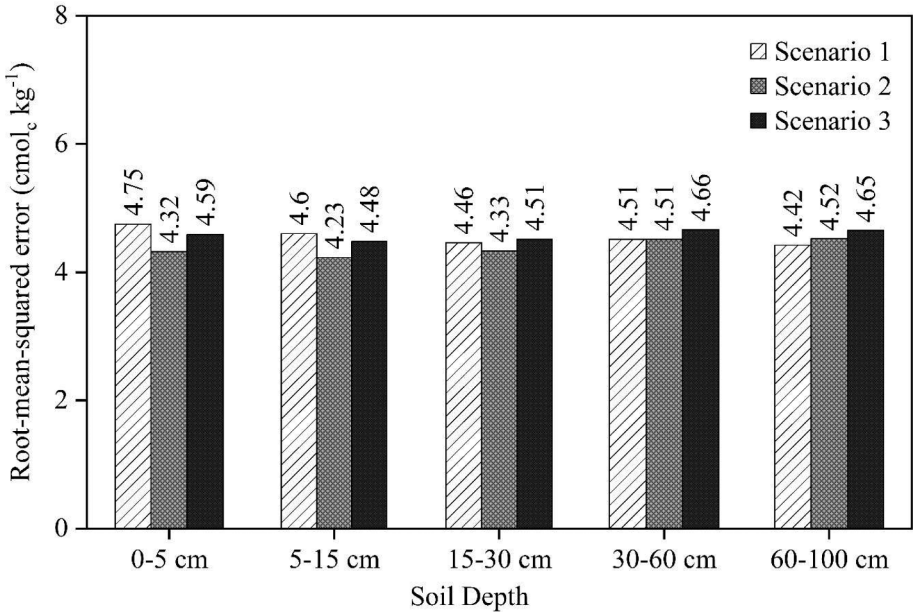


Fig. 6. Root-mean-squared errors (RMSE) of DSM model for the three scenarios at six different soil depths.

% among different soil layers across 20 different AERs. When averaged over all the six GlobalSoilMap-specified depths, the depth-weighted θ_{FC} and $\boldsymbol{\theta_{PWP}}$ values ranged from 17.1 % and 7.5 % (AER 2) to 34.3 % and 19.8 % (AER 6), respectively. These values were slightly lower than the averages estimated from the harmonized data for the 1707 profiles (θ_{FC} and θ_{PWP} values of 38 % and 21.8 % for the AER 6, respectively). A depth-wise comparison between θ_{FC} data from the harmonized profile data and DSM-estimated average θ_{FC} data for different AERs showed strong positive correlation (*r* values ranged from 0.83 for the 60–100 cm soil layers to 0.89 for 0–5 cm soil layers) suggesting close match between such averages. Similar correlations were also found for the harmonized and DSM-estimated θ_{PWP} values across 20 AERs. Between the two extremes of Western Plains and Deccan Plateau, almost the entire northern half of India (AER 2, AER 1, AER 4, AER 16, AER 14 and AER 13) had the depth-weighted θ_{FC} averages below 25 % and θ_{PWP} averages below 12.5 % except for the Northern Plains (AER 9). This large geographical area (~130 million ha) includes one of the most intensively cultivated regions of the country with highly developed irrigation infrastructure (Shah et al., 2021) to the tune that Green Revolution is often questioned for its ill impacts of soil salinity and land degradation. On the other hand, a large tract of Deccan Plateau (AER 6, AER 3, and AER 7), Eastern Coastal Plains (AER 18), and part of Central Highland (AER 5) are bestowed with high plant-available water contents ($\theta_{FC} - \boldsymbol{\theta_{PWP}}$) exceeding 13 %.

Linear and nonlinear dependency analyses using Pearson and distance correlation measures show that sand content is the dominating basic soil property influencing the θ_{FC} with *r* values ranging from −0.62 to −0.91 (dCor: 0.58 to 0.87) followed by clay contents (*r*: 0.33 to 0.87; dCor: 0.36 to 0.86), SOC contents (*r*: 0.22 to 0.52; dCor: 0.39 to 0.62) and soil pH (*r*: −0.01 to −0.33; dCor: 0.39 to 0.46). Except for having a slightly stronger correlation than θ_{FC}, the dependency measures for θ_{PWP} followed the same order as for θ_{FC}: sand contents (*r*: −0.60 to −0.85; dCor: 0.57 to 0.81) > clay contents (*r*: 0.40 to 0.86; dCor: 0.39 to 0.86) > SOC contents (*r*: 0.46 to 0.76; dCor: 0.47 to 0.82) > soil pH (*r*: −0.23 to −0.56; dCor: 0.33 to 0.62). Stronger influences of sand and SOC contents on water retention capacities (θ_{FC} and θ_{PWP}) in contrast to the influence of clay contents and soil pH on the nutrient retention capacity (CEC) are consistent with the well-known facts that soil structure majorly influences soil water relationships while charge characteristics (permanent and pH-dependent) influences cation exchange behavior in soil. Soils with high SOC and clay contents are known to have large and stable soil aggregates (Bronick and Lal, 2005; Purushothaman et al., 2022); soils with such stable aggregates are expected to have high available water contents. Thus, both θ_{FC} and θ_{PWP} are expected to be positively correlated to both clay and SOC contents because of their role in the formation and stabilization of soil aggregates. With moderate to strong weathering intensities, soil pH values < 7 were observed in 3681 out of 7317 soil layers (~50 %) used in this study suggesting the presence of iron and aluminum oxides and hydroxides, which contributes to soil aggregation (Bronick and Lal, 2005; Sarathjith et al., 2014; Vasava et al., 2019). Negative correlation coefficients between soil pH and water holding capacities may be a result of low pH-mediated influence of Fe- and Al- oxides and hydroxides on soil aggregation.

3.4. Relationship between secondary soil properties and environmental covariates

Among environmental covariates, the terrain attribute of valley depth showed strongest negative correlation coefficients with layer-wise CEC values (*r*: −0.68 for 5 cm soil depth to −0.72 for 100 cm soil depth) followed by the mean monthly rainfall during April (*r*: −0.43 for 5 cm depth to −0.52 for 100 cm depth), mean monthly EVI for May (*r*: −0.39 for 5 cm depth to −0.45 for 100 cm depth), and mean monthly temperature during May (*r*: 0.36 for 5 cm depth to 0.44 for 100 cm depth). This correlation structure slightly changed when dCor was used as a nonlinear dependency measure. Similar to *r*, valley depth also emerged as the strongest environmental covariate for CEC followed by the monthly rainfall during April (dCor: 0.47 for 5 cm depth to 0.55 for 100 cm depth), monthly EVI for May (dCor: 0.53 for 5 cm depth to 0.56 for 100 cm depth), and temperature during March (dCor: 0.51 for 5 cm depth to 0.55 for 100 cm depth). Valley depth was also negatively correlated with AER-averaged clay contents; the negative correlation between post-monsoon rain (e.g., April rainfall) and positive correlation with May temperature are characteristic climatic conditions favorable for clay illuviation. Thus, CEC values in Indian soils are primarily driven by the pedogenesis of clays in which monsoon rain favors clay translocation to deeper soil layers followed by the post-monsoon dry weathers favoring subsequent deposition of clay contents in B horizons. The dependency measures for θ_{FC} and θ_{PWP} suggest that the October rainfall and January temperature are the most influencing environmental covariates. For θ_{FC}, January temperature (*r*: 0.08 to 0.62), October rainfall (*r*: 0.24 to 0.40), elevation (*r*: 0.00 to −0.44), and April

EVI (r: 0.22 to 0.26) were most influencing covariates. Similarly, October rainfall (r: 0.49 to 0.58), January temperature (r: 0.14 to 0.62), and April EVI (r: 0.44 to 0.48) were most influencing covariates for θ_{PWP}. No terrain variable was selected for the θ_{PWP} in the feature selection step. Corresponding dCor values also suggested the similar trend with rainfall and temperature being the critical environmental covariates influencing water holding capacities. Interestingly, the NDVI value was observed to be an influencing covariate for θ_{FC} based on the mean dCor value (dCor: 0.45 15 cm soil depth to 0.66 for 100 cm soil depth). Thus, correlation analysis using 20 different environmental covariates show that nonlinear relationships dominate for both nutrient and water holding capacities with shallow valley depth generally favoring high CEC (highest dCor value of 0.75) while high January temperature promoting high θ_{FC} (highest dCor = 0.67) and θ_{PWP} (highest dCor = 0.68) values in Indian soils. Out of vegetation indices, the EVI values during the summer months (April to June) were frequently observed to influence CEC and water holding capacities.

Malone et al. (2020) also have observed bioclimatic variables significantly influencing water holding capacities. Through a hybrid feature selection, these authors showed that temperature during the wettest and driest quarter and rainfall during the warmest quarter are most significant covariates for soil hydraulic properties. January in India coincides with the mid-winter season (December to February) with the lowest annual specific humidity (Jaswal and Koppar, 2011). Thus, the temperature during the driest quarter of the year appears to be a significant driver for the evolution of soil hydraulic characteristics in both these subcontinents reaffirming the role of *scorpan* framework on soil attributes such as soil hydraulic characteristics.

3.5. Relationships between soil properties and agronomic attributes

Soil function parameters are some of the final products of DSM, which can be used for managing agricultural resources (Sanchez et al., 2009). Although we had low soil profile density, we examined the linkages between soil properties (basic soil properties from Reddy et al., 2021 and the secondary soil properties of CEC and AWC) and agronomic attributes. We estimated Pearson correlation coefficient to relate profile-averaged soil properties with state-wise average agronomic attributes. Except for CI and grain yield, all the remaining seven agronomic attributes shows positive correlation with CEC and AWC values (Fig. 7A). Specifically, fertilizer consumption and both the gross and net sown areas were significantly influenced by CEC ($p < 0.01$). For AWC, the fertilizer consumption was the only significantly correlated agronomic attribute ($p < 0.01$). Positive correlation coefficients with the agronomic attributes suggest that farmers tend to use more area for cultivation with greater irrigation intensities and, perhaps, take the advantage of applying more fertilizers when a soil has high water and nutrient holding capacities. This may also be seen with linearly increasing NSA values with increasing CEC (Fig. 7B). Fig. 7C shows that the irrigation intensities are generally high for the sandy soils; the reverse was observed for the clayey soils. The negative correlation between yield and water and nutrient holding capacities was also observed when we estimated correlation coefficients for each soil layer. Although such a result is not expected, Fig. 7D shows that the grain yield bears a parabolic relationship with SOC contents: higher grain yield at lower SOC contents eventually increasing after SOC content increase beyond the threshold value of 0.4 %. Higher yield for low SOC soils is clearly associated with the high consumption of NPK fertilizers for corresponding Indian states.

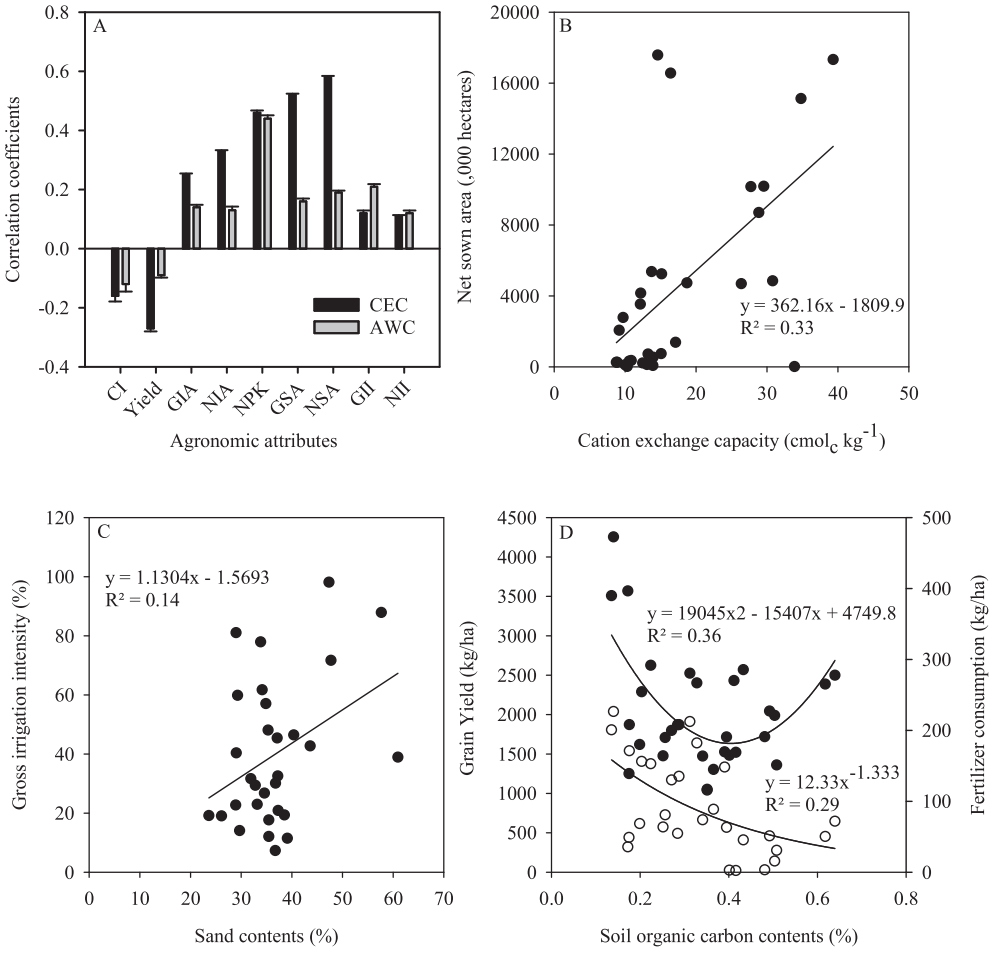


Fig. 7. Pearson correlation coefficients (A) between agronomic attributes and cation exchange capacity (CEC) and plant available water content (AWC) and scatter plots between (B) CEC and net sown area, (C) sand content and gross irrigation intensity and (D) between soil organic carbon contents and grain yield and NPK fertilizer consumption. All agronomic attributes were obtained by averaging their state-averaged annual estimates over 2004 to 2017; all soil attributes were obtained by averaging the digital soil mapping-estimated values over a state to a depth of 2 *m*.

Reddy et al. (2021) showed distinct positive associations between clay and SOC contents. Soils with high clay and high SOC contents typically have high CEC values. Although not significant even at p value <0.05, the negative correlation between CEC and grain yield may be because of the high NPK use in low SOC soils. Our results are based on soil and agronomic attributes averaged over large areas (state level) and, therefore, provide a broad view on how secondary soil properties influence agronomic attributes. Nevertheless, these results reinforce the assertion that water holding capacities can play a major role in agricultural management activities. Specifically, indiscriminate use of water without the capacities of a soil profile to store water should be practiced (Sanchez et al., 2009).

4. Conclusion

Legacy soil data and measured soil properties from different sources were combined with terrain and bioclimatic variables and vegetation indices using machine learning and digital soil mapping approaches to develop digital soil maps for water and nutrient holding capacities for Indian soils for six GlobalSoilMap-specified depths at 500 m resolution. Original legacy soil data were available for 1681 locations for CEC across the country while about a third of these locations had θ_{FC} and θ_{PWP} data. Because RF models and bioclimatic factors have shown promise to fill such data gaps, we used a machine learning framework to create national-scale soil maps for these secondary soil properties. Similar to many DSM studies, estimated R^2 values in the 10-fold cross-validated datasets ranged from 0.60 to 0.65 for CEC while a substantially lower accuracies were found for the water holding capacities (R^2 for θ_{FC}: 0.27–0.34 and R^2 for θ_{PWP}: 0.33–0.37). Resulting soil function maps show distinct spatial patterns that are closely linked to the parent material and bioclimatic variables consistent with the *scorpan* framework. Soils developed from basaltic parent materials of central and part of peninsular India had high CEC and water holding capacities while opposite trends were seen in soils developed in the deserts of Western Plains and in soils developed from granite-gneiss parent material commonly seen in many eastern Indian regions. Terrain attributes were also observed to have stronger influence on the distribution (and, possibly genesis) of these secondary soil properties than the climatic variables. Specifically, valley depth showed strongest negative correlation coefficients with layer-wise CEC values (r: −0.68 to −0.73) while January temperature (the driest month of the year) showed strongest influence on water holding capacities (r: 0.08 to 0.62 with θ_{FC} and 0.14 to 0.62 for θ_{PWP}). October rainfall also appeared to influence θ_{PWP} with the correlation coefficient of 0.49 to 0.58. Both linear and nonlinear correlation analyses showed that the occurrence of high CEC soils are also linked to summer (April and May) temperature. Our analysis suggests that CEC in Indian soils is primarily driven by the pedogenesis of clays in which monsoon rain favors clay translocation to deeper soil layers followed by the post-monsoon dry weathers favoring subsequent deposition of clay contents in subsurface horizons. Similarly, strong linkages between January temperature and water holding capacities reaffirms the conclusion that temperature of driest quarters is a key driver of water relationships in soil as was also observed in the Australian soils by Malone et al. (2020). Analysis of the linkages between DSM-estimated soil properties and selected agronomic attributes suggest that both CEC and AWC values are positively corelated with the gross and net sown area and area under irrigation. A strong positive correlation between CEC and NPK fertilizer consumption and generally positive correlation between water and nutrient holding capacities with irrigation intensities were also observed suggesting that soils with high CEC and AWC tend to encourage farmers to invest more on both water and nutrients.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.geoderma.2022.116265.

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